上海交通大學
SHANGHAI JIAO TONG UNIVERSITY

# TC3000J
# Technical Communication

## Introduction to Technical Communication

**Dr. Xi Liu**

*Adjunct Lecturer*

Global College, Shanghai Jiao Tong University, Shanghai, China

# Course Overview

# Course Info

| | | |
|---|---|---|
| **Lecture Time:** WK1 –WK8 | Section 1 | Weds 10:00 – 11:40 |
| | Section 2 | Weds 12:10 – 13:50 |
| | Section 3 | Weds 16:00 – 17:40 |
| **Location:** | DXY-311 | |
| **Instructor:** | Dr. Xi Liu | |
| **Office:** | Room 407B | Longbin Building |
| **Office Hour** | Weds or online | By appointment – min. 3 days in advance |

## Notifications

Canvas

## Discussions

Feishu

# Course Outcomes

1) **Learn the methodology to define an engineering problem** via logically analysing needs, solutions, and boundary conditions.
2) **Paraphrase and directly quote** references from academic journals and trade magazines, with appropriate citation formatting
3) **Create** a variety of **technical charts and diagrams**, based in the type of data being reported
4) **Integrate charts, diagrams, and equations into a report**, with appropriate signalling words to highlight key points in the visuals
5) **Deliver a short technical presentation**, with appropriate structure, intonation, and body language
6) **Write a professional e-mail**, with tone and structure that is appropriate for purpose and audience
7) Craft a **resume for employment** or graduate school, with objective descriptions of academic and professional achievements
8) Craft a **cover letter for employment** or **statement of purpose for graduate school**

# Schedule

| Lecture | Date | Lecture Topic |
|---|---|---|
| 1 | 2026.03. 04 | Introduction To Technical Communication |
| 2 | 2026.03.11 | Engineering Design and Technical Documentation |
| 3 | 2026.03.18 | Quotation and Paraphrasing |
| 4 | 2026.03.25 | Professional Correspondence<br>Resume and Cover Letters |
| 5 | 2026.04.01 | Visual Aids |
| 6 | 2026.04.08 | Public Speaking |
| 7 | 2026.04.15 | Mock Presentations (Practice)/Final Presentations (Grading) |
| 8 | 2026.04.22 | Final Presentations (Grading) |

# Course Description

The key areas covered by the course will include but not limited to:

- Technical project design through a **problem-need-solution-validation framework**
- **Technical diagrams and data visualization**, and their use in the reporting of results
- **Workplace-related communications** such as CVs and professional correspondence
- Paraphrasing, direct quotation, and citation formatting
- Primary features and linguistic structures of technical language

# Major Assignments

**Homework Technical Project Design** — 25% of total

- Technical Project Design
  30 Pts

**Homework Technical Description** — 20% of total

- Technical Description
  20 Pts

**Homework Project Presentation** — 30% of total

- Project Presentation
  30 Pts

# Assignments and Grading Policy

| Assignments | Grading Policy |
|---|---|
| • Technical Project Design | 25% |
| • Data Presentation and Description | 20% |
| • Project Presentation | 30% |
| • Attendance | 5% |
| • Participation | 20% |
| Course Evaluation | 1% Bonus |

- Assignments published on CANVAS
- Submission on CANVAS
- **Late submissions for homework:** -20%, if ≤24 hrs; -30%, if ≤48hrs; **-100%, if later**.
- Grades not changed at the end of the semester, no recovery of lost points

# In-Class Participation

- **Active participation** is expected and evaluated.
- Students are not allowed
  - Use their smartphones
  - Do anything that disturbs others
- Repeated violations may lead to points deductions or confiscation of smartphone until the end of the class.
- In-class assignments count **towards participation credits**.
- To count for credit, assignments must be submitted when demanded and the effort to finish as much as possible must be evident.
- **Attendance** is a  prerequisite for in-class assignments
- Teamwork will be graded via peer review process.

# Information Security

It is forbidden to share course materials or login information (e.g., usernames, passwords, meeting IDs) to any person not enrolled in the course during this semester. Course materials may never be made accessible to any person not affiliated with the GC SJTU. Course materials must never be shared online. Students may not share their classwork or homework with any person – enrolled in the class or not – until the semester is finished. The abovementioned rules still apply, meaning that classwork and homework may never be shared outside the UM-SJTU JI or posted anywhere on the internet or on social media.

# Honour Policies

- Assignments must be completed independently, unless otherwise stated (e.g. group work)
- No work from your previous courses may be used
- Plagiarism is **NOT** accepted
- Discussion on a conceptual level is allowed
  - Calculations and writing must be completed independently
- Violations face harsh penalties
  - Warnings, loss of points, report to Honour Council
- Honour Council may impose further penalties
  - up to expulsion from GC or SJTU

# Plagiarism and the Honour Council

## What is plagiarism?

The use of three or more words of any source text (academic or otherwise, including written by you) without the appropriate paraphrasing/quotation and citation.

## Depending on

- Severity of violation
- Frequency of violations

## Consequences (at instructors discretion)

- Warning
- Partial or complete deduction of points on that assignment
- Reporting the violation to the Honor Council

# Honour Policies

- The use of AI tools is for now permitted for TC3000J in this section, but terms **must be obeyed**:
  - Their contribution **must be clearly referenced**. In particular, the prompts that were given by an AI-based tool must be highlighted and acknowledged. One must specify how the tool was used, e.g. how the output was modified.
  - **Students are liable for any AI-generated contents** they use for any course work or exam.
  - **Under no circumstances may AI-generated contents be copied without critical analysis, nor used to fabricate data, sources, or experiments.**
  - The use of AI-based tools may be limited or prohibited for certain assignments at the instructor's discretion.

# Internet Resources

- CANVAS will be used extensively
- All material will be published on CANVAS
- Make sure to check CANVAS daily

- Use of Discussion and Chat Room functions is encouraged
  - Don’t post solutions to any assignments there => Honour Code violation

# Feel free to…

Interrupt me during the class and **ask questions** related to the TC topic in discussion!

Let me know when my teaching is too slow or too fast, too easy or too difficult!

Let me know when you have special interest in a particular topic!

# You?

Please prepare a brief introduction about yourself. (Name, Major, Study Interests and Plans, Expectations, Your strong points, …)

In-class assignment of Lecture 1

Assignments › Lecture 1: Introduce Yourself and Make Profile…

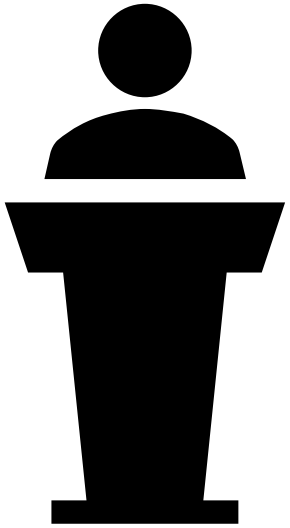
## Lecture 1: Introduce Yourself and Make Profiles of Others

Published Edit

1. Please introduce yourself in your team via a small oral presentation. If you have not found yourself a group before the beginning of this assignment, you are required to introduce yourself in front of the class. You have maximum min. (1 point)

2. Make profiles of the other four members of your group and submit the profile online. Please use the table template below. (1 point)

| Team n | Profile no.: | Name: |
|---|---|---|
| Major | | |
| Characters | | |
| Strong points | | |
| To be improved | | |
| What he/she wants to bring | | |
| Others … (you can also add rows below if needed) | | |



# Introduction
# to
# Technical Communication

# Technical Communication

- **The ultimate goal of communications:** Getting the message cross!

- **Steps of Communication**

  Sender: Idea formation → Message encoding
  Channel: Message transmission
  Receiver: Message decoding → Feedback

- **Purpose**

  To promptly and appropriately have the technical tasks planned; the data generated and collected; information released, indexed, monitored, and controlled.

- **Not trivial**

  1. Natural hindrance/"noise" in communications
  2. Tasks are important thus standards are high (Technical)

- **Judgment** What can be counted as effective communications?

  The information can be transferred to the *right* audience, in the *right* way, at the *right* time, so that the *right* impact can be made among the audience.

M.J.Wilson et. al., Sending the right message to the right audience. FACETS 1: 127–137. doi:10.1139/facets-2016-0015

# Objective Hindering Factors

- Environment Factors
- Information Overload
- Knowledge (e.g. Jargon / Terminology) Deficit
- Habitual Differences
- Selective Perceptions
- Multi Channels
- Bad Attitude
- Emotion

# Improvement Strategies

- **The ultimate goal of communications:** Getting the message cross!

- **Types of Communication**
  Commonly verbal, nonverbal, written, or visual interactions

- **Judgment** What can be counted as effective communications?
  The information can be transferred to the *right* audience, in the *right* way, at the *right* time, so that the *right* impact can be made among the audience.

- **What Good Tech Comm Must Take Care of?**
  Purpose, audience, content, format, process

- **Features of Good Tech Comm**
  Accurate
  Coherent
  Clear
  Concise
  Appropriate

M.J.Wilson et. al., Sending the right message to the right audience. FACETS 1: 127–137. doi:10.1139/facets-2016-0015
NC State University, Technical Communication Videos, https://www.youtube.com/playlist?list=PLR-UWtI0r_GDI8SOgsyTR4rmW_gtprqB6

# Coming Next Lecture

**Engineering Design**

- Problem - Need - Solution - Validation Framework

**Technical Documentation**

- What types of documents are commonly seen for technical communication?
- How to compose a technical document?
- Important elements in a technical document.

**Introduction and discussion by students**

# Preparation for Future Lectures

- **Familiarize with technical documentation!**

Find of three types of technical documents that you can imagine that you might encounter according to your education and career plan.  (Please complete individually). Deadline: before lecture 2.

- **Group up!**

Find your group members that you would like to work with towards the final project presentation. Deadline: TODAY!

- **Preparation for homework assignment 1.**

Start with finding an engineering problem that you would like to work on. Homework details will be released in Lecture 2; students' ideas are expected for discussion in class; final submission before lecture 5.

**See you next time!**

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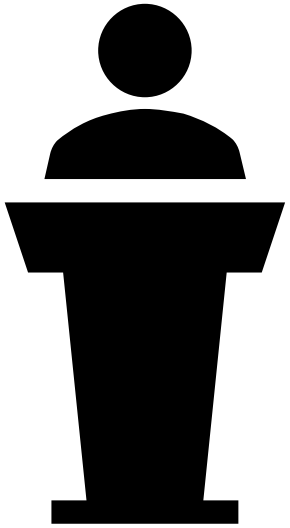
# Student Time

Group Up!

Introduction speech

Profiling